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UAV Formation Flight Control Method Based on DQN

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Abstract: Currently, unmanned aerial vehicle (UAV) formation flight control still relies on model - based information, and the level of UAV intelligent control is relatively low. To address this issue, a deep reinforcement learning approach is investigated.Firstly, corresponding reinforcement learning elements are devised for formation control problems, and formation controllers based on the deep reinforcement learning Deep Q-Network (DQN) algorithm are designed. Simultaneously, a method integrating a priority strategy with a multi - layer action library is proposed to accelerate the algorithm convergence and enable the wingman to ultimately maintain the expected range.Finally, the designed controller is compared with the Proportional - Integral - Derivative (PID) controller through simulation, and the effectiveness of the DQN controller is verified. The simulation results indicate that the controller can be applied to UAV formation, enhance the intelligence of the wingman, maintain the expected distance through autonomous learning, and the controller design does not require accurate model information, which provides a basis and reference for the intelligent control of UAV formation.
Keywords: formation flying; reinforcement learning; intelligent control; DQN

Article Highlights:

Integrates DQN with UAV dynamics to create a specialized structure for formation control.

PyTorch-based simulation with tensor computing accelerates training, enabling rapid iteration and optimization of control strategies.

Comparative analysis shows DQN outperforms PID and DDPG in accuracy and stability, making it ideal for UAV formation control.

1. Introduction

UAVs have been extensively utilized in military and civilian domains owing to their numerous characteristics, including light weight, low cost, and excellent concealment. On the basis of individual UAVs, UAV formation flying has significantly broadened the scope of UAV applications. This has emerged as the development trend of UAV technology in recent years and has been widely recognized by countries around the world.As the requirements of various military and civilian missions continue to increase and the complexity of these missions grows, the performance of a single UAV has become insufficient to meet both civilian and military needs. Through the study of swarm phenomena in organisms such as ants and bees, researchers have discovered that the cooperative behavior of biological colonies exhibits superior performance in a variety of complex tasks. By applying this concept of swarm intelligence to UAVs, the formation control technology of UAVs has been developed.In recent decades, the formation control technology of UAVs has been successfully implemented in various military and civilian missions. Consequently, the formation control technology of UAVs has become a research focus. [1].

Currently, substantial achievements have been attained in the formation control technology of unmanned aerial vehicles. The primary formation control structures in the existing literature encompass the navigator - following method, the virtual structure method, the behavior - based method, the graph - theory - based method, and the consistency theory [2-6]. Among these, the navigator - following method is the most prevalently employed approach at present, owing to its merits of structural stability and ease of implementation.Regarding control methods, domestic and foreign scholars have also applied PID and backstepping control to formation control and achieved corresponding outcomes[7]. The design of the PID controller is straightforward and does not necessitate model information; however, issues such as parameter adjustment still persist. Backstepping, as a classical nonlinear control method, yields favorable results in formation control. Nevertheless, it typically demands accurate models for controller design, which is challenging in practical applications. Simultaneously, these controllers lack sufficient intelligence to meet the current requirements of intelligent mission execution; hence, novel control methods need to be developed.

Reinforcement learning (RL), a sub - field of machine learning, has witnessed rapid development in recent years and has been extensively applied in game theory, optimal scheduling, and robot control. Its fundamental principle is that when a task conforms to the Markovian Decision Process (MDP), the training agent aims to maximize the reward through interaction with the environment, ultimately learning the optimal strategy. The merit of this approach lies in its ability to autonomously learn the best strategy without prior knowledge of the exact model of the environment.Currently, the reinforcement learning method has been utilized in the domain of UAV control, such as path planning, collaborative decision - making, and single - machine control[7-8]. It has also been applied to the formation control of UAVs[9]. A Q-learning algorithm with a dynamically changing learning rate has been designed to achieve the clustering of fixed - wing UAVs in a random environment. Some literature takes

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three UAVs as an example and uses the DDPG algorithm to implement clustering for navigation. A CACER algorithm based on the Actor - critic framework has been proposed to address the leader - wingman aggregation problem in a continuous state[9-10]. Certain literature employs the Q-Learning approach, enabling wingmen to learn to aggregate within a specific distance from a lead aircraft in a static random environment. Some literature has proposed a cluster reinforcement learning method to group UAVs into the desired formations at target points[11-13].

Notable advancements have been achieved in both theoretical investigations and practical implementations of UAV formation control technology. Nevertheless, certain challenges and limitations persist and necessitate resolution.Firstly, extant formation control approaches frequently rely on accurate models or simplified suppositions, which may not be valid in intricate and dynamic settings. This restricts their adaptability and robustness in real - world situations. Secondly, current research predominantly centers on the formation control of a small number of UAVs, and the scalability of these methods for large - scale UAV formations remains a crucial concern. As the quantity of UAVs in the formation rises, the complexity of control problems escalates exponentially, rendering it arduous to attain stable and efficient formation control. Thirdly, the intelligence quotient of existing formation control methods is still relatively low. Therefore, the DQN, as an advanced approach integrating deep learning and reinforcement learning, possesses potent self - learning and decision - making capabilities. It can approximate the Q-function via deep neural networks to manage high-dimensional state spaces without depending on accurate environmental models. In UAV formation control, DQN empowers each UAV to function as an intelligent agent, continuously learning the optimal control strategy through interaction with the environment, so as to achieve stable flight of the formation, task coordination, and respond to diverse complex environments and emergencies. The incorporation of DQN is anticipated to substantially enhance the adaptability, robustness, and intelligence level of UAV formation control, offering more reliable technical support for the application of UAVs in complex tasks.

2. Description of UAV Formation Control Problems

During formation flight, relative motion exists between any two UAVs. The formation mode considered in this paper is the "leader-wingman" mode. Thus, the mathematical model of multiple UAV formations can be derived by analyzing UAVs in two - unit formations. This section takes two UAVs in formation flight as an example. The experimental coordinate system of the UAV is selected as the reference frame for research, with the leader and the wingman as the specific cases. Assuming that the track azimuth of the UAV is approximately equivalent to the heading angle, in all subsequent discussions, they are regarded as equal. That is, the heading angle can be used instead of the track azimuth in expressions. The relative motion of the two UAVs is presented in Figure 1.

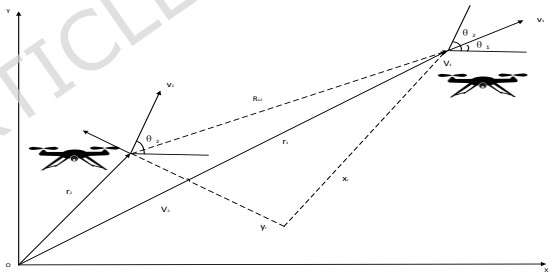

Figure. 1 Schematic diagram of relative motion of UAVs

According to literature, the equations of relative motion for UAV formation flight are as follows:

$$\begin{cases} \dot{X}_i = v\cos\theta_i \\ \dot{Y}_i = v\sin\theta_i \\ \ddot{\theta}_i = f(\theta_i, \theta_{ic}) \end{cases} \tag{1}$$

Where: θ_1, θ_2 --Heading angle (track azimuth) of UAVs V_1 and V_2;

v_1, v_2 --Speed Scales for UAVs V_1 and V_2;

R_{12} --Relative position vector of UAVs V_1 and V_2;

r_1, r_2 --Position vector of UAVs V_1 and V_2 in the ground coordinate system.

The objective of this study was to enable a wingman to maintain a consistent speed at varying altitudes while taking into account the formation distance in the y - direction. Moreover, the wingman should be capable of maintaining the desired formation distance by learning to adjust its own heading angle, as depicted in Figure 1. To attain the aforementioned objectives, the currently prevalent design methods for formation controllers frequently rely on model - based design. However, in practical scenarios, the modeling process is influenced by

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numerous factors. Simultaneously, UAVs exhibit relatively low intelligence and lack autonomous learning capabilities. Therefore, reinforcement learning is employed to address this issue.

3. Design of Formation Controller Based on DQN

3.1 Reinforcement learning and DQN algorithm

Figure 2 depicts a theoretical model of reinforcement learning. It is assumed that there exists an individual within the environment capable of implementing action strategies, referred to as an agent. At time t, the agent perceives the state information s_t of the current environment and generates the action at via the policy. The agent executes an action to act upon the environment and transitions to a new state s_{t+1}. Simultaneously, it receives the reward signal $r(s_t, a_t)$ fed back by the environment, thereby initiating a new cycle. Agents amass experience, modulate strategies, and ultimately acquire the optimal action strategy via continuous interactive trial - and - error with the environment, aiming to attain the maximum cumulative reward upon task completion.

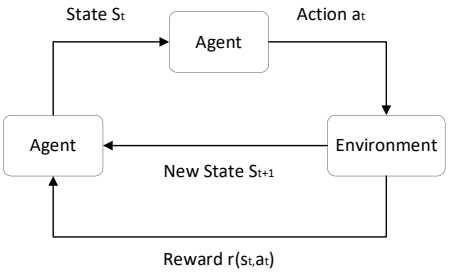

Figure. 2 Theoretical Framework for Reinforcement Learning

Deep Q-Network (DQN) employs a deep neural network to approximate the Q-value function, taking states as the input and the output corresponding to the Q-value of each action. Let the parameter of the neural network be θ; then, the Q-valued function can be represented as $Q(s, a; \theta)$. Network parameters are optimized through the minimization of the loss function, which is defined as the mean square error between the predicted Q-value and the target Q-value:

$$L(\theta) = E_{(s,\alpha,r,s')\sim D}[(y_i - Q(s, a; \theta))^2] \tag{2}$$

Where: the target Q value y_t is calculated as:

$$y_t = r_{t+1} + \gamma \max_{a'} Q(s', \alpha'; \theta^-) \tag{3}$$

Where: γ is the discount coefficient θ^- is the parameter of the target network, copied periodically from the policy network.

3.2 Design of key elements

The intelligent agent engages in interaction with the environment to ultimately acquire the optimal strategy. The environment is composed of state space, action space, and reward function. In this section, the key elements in the interaction process of the formation control problem are designed sequentially.

State space: Based on the aforementioned relative kinematic model of the drone, it is evident that the distance in the y-direction and the relative heading angle between the leader and the wingman can be transmitted via a communication link. This information can serve as the state space. The state space S is defined as:

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$$s = \begin{cases} s_1 = e_y \\ s_2 = \dot{e}_y \\ s_3 = \int e_y \\ s_4 = y \\ s_5 = e_\theta \\ s_6 = \dot{e}_\theta \\ s_7 = \int e_\theta \\ s_8 = \theta_F \end{cases} \tag{4}$$

Where: $e_\theta = \theta_L - \theta_F$ represents the relative heading angle between the leader and the follower; $e_y = y_d - y$ denotes the error between the desired y-direction distance and the actual y-direction distance; y_d represents the desired y-direction distance. Based on the PID control concept, the integral and differential terms of e_y and e_θ are incorporated into the state space, enabling the formation controller to better learn the strategy.

Hierarchical action space: In the process of formation flight, the wingman sustains the desired distance in the y - direction through the adjustment of the heading angle. Consequently, the action space is defined as the desired heading angle command of the wingman. The action is updated by the controller after each sampling, and the command remains invariant during the sampling period. Given that the employed DQN is a discrete algorithm, the desired command necessitates discretization. With regard to control accuracy, the hierarchical heading angle action library is designed as follows:

$$action_1 = [-\varphi_{\max} \cdots -20, -10, 0, 10, 20 \cdots \varphi_{\max}] \tag{5}$$

$$action_2 = [\frac{-\varphi_{\max}}{10} \cdots -2, -1, 0, 1, 2 \cdots \frac{\varphi_{\max}}{10}] \tag{6}$$

$$action_3 = [\frac{-\varphi_{\max}}{50} \cdots -0.4, -0.2, 0, 0.2, 0.4 \cdots \frac{\varphi_{\max}}{50}] \tag{7}$$

In the formula, $[-\varphi_{\max}, \varphi_{\max}]$ represents the range of heading angles, where φ <0 indicates that the drone is deviating to the left. i=1,2,3 denotes three different sets of actions, each with a different discrete interval. The purpose of designing different action sets is to enable the controller to select different action sets to output corresponding desired commands under different situations. For example, when the distance error is large, a larger action set with a larger spacing is selected to quickly reduce the distance error. When the distance error decreases, a smaller action set is selected for fine adjustment, which can effectively improve control accuracy.

Reward Function: The formation control problem of unmanned aerial vehicles represents a typical sparse-reward problem. In this context, the controlled entity encounters difficulties in obtaining rewards at the initial stage and necessitates a series of explorations and intricate operations to acquire them. Consequently, the design of the reward function is closely associated with the ultimate training outcomes. Simultaneously, the formation control problem diverges from other environments in that the intelligent agent does not halt upon attaining the maximum reward but persists in forward flight. The designed reward function should be as simplistic as feasible, which can empower the wingman to learn the optimal strategy while possessing strong transferability and low computational complexity. The designed reward function is:

$$\begin{cases} r_1 = \begin{cases} 5 & \text{if } |e_y| \le 0.1 \\ 0 & \text{otherwise} \end{cases} \\ r_2 = -0.001 * |e_y| \\ r_3 = \begin{cases} -50 & \text{if done} \\ 0 & \text{otherwise} \end{cases} \\ R = r_1 + r_2 + r_3 \end{cases} \tag{8}$$

In the formula: r_1 is the reward, which is obtained when the distance error is reached; r_2 is the punishment, and the size of the punishment depends on the distance. The larger the error, the greater the punishment, and vice

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versa, the smaller it is, with a minimum of 0; r_3 is the early stop penalty, which is obtained when the training process is stopped prematurely or unexpectedly; R is the sum of all rewards and punishments obtained during the process. The value of r_1 represents the ultimate control objective, while the functions of r_2 and r_3 are to guide the strategy towards obtaining rewards. The maximum penalty term of r3 is to avoid reaching the end condition as much as possible, while $r_1 > r_2$ makes the final reward positive.

Termination condition: In the training process, the validation duration for each round is Td. During the flight, if the distance in the y - direction is either excessively large or extremely small, it is unfavorable for formation flight. When the distance is overly large, it may augment communication complexity or even result in loss of contact. When the distance is too small, it may give rise to collisions. Consequently, the termination condition for the training process is established as follows:

$$\begin{cases} 1 & \text{if } y < y_{min} \ or \ y > y_{max} \text{ or } T = T_d \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

Where: $[y_{min}, y_{max}]$ is the distance range set by the formation.

3.3 Design of Formation Flight Controller

The research aim of this section is to devise a formation controller founded on the Deep Q-Network (DQN) to generate the heading angle command for the wingman. This enables the wingman to autonomously acquire the ability to trail the leader and uphold the desired distance without any pre-existing knowledge. The control framework is presented in Figure 3 below. In this figure, the drone formation comprises two wingmen, possessing identical state spaces, action spaces, and rewards as those formulated in the preceding section.

The controller segment encompasses a Memory unit, a DQN network, and an action selection strategy component. The Memory unit serves to store data from the interaction process and conducts random sampling at fixed time intervals to extract batch samples for the purpose of updating the DQN network. The neural network architecture outputs all action values, and subsequent to this, the action selection strategy generates the wingman's heading command. The wingman reacts to this command, modifies the state of the leader, accomplishes the state update, and computes the current reward in accordance with Equation (8). The reward and state are then fed back to the controller jointly, thereby establishing a comprehensive formation control structure.

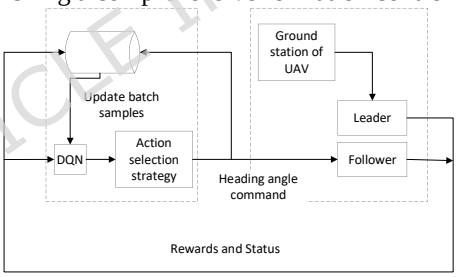

Figure. 3 Control Structure of DQN Formation

The pseudo code of the algorithm is shown in Table 1:

Table. 1 Pseudo code Program of DQN

Algorithm 1 Deep Q-learning with Experience Replay
Initalize replay memory D to capacity N
Initialize action-value function Q with random weights θ
Initialize target action-value function $\hat{Q}$ with weights $\theta^{-} = \theta$
For episode=1,*M* do
Initialize sequence $s_1 = \{x_1\}$ and preprocessed sequence $\phi_1 = \phi(s_1)$
For t=1,T do
With probability ε select a random action a_t
Otherwise select $a_t = \arg\max_a Q(\phi(s_t), a; \theta)$
Execute action a_t in emulator and observe reward r_t and image $x_t + 1$

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Set $s_{t+1} = s_t$, a_t, $x_t + 1$ and preprocess $\phi_{t+1} = \phi(s_{t+1})$
Store transition $(\phi_t, a_t, r_t, \phi_{t+1})$ in D
Sample random minibatch of transitions $(\phi_j, a_j, r_j, \phi_{j+1})$ from D
Set $y_j = \begin{cases} r_j & \text{if episode terminates at step j+1} \\ r_j + \gamma \max_{a'} \hat{Q}(\phi_{j+1}, a'; \theta^-) & \text{otherwise} \end{cases}$
Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ with respect to the network parameters θ
Every C steps reset $\hat{Q} = Q$
End for
End for

3.4 Priority selection strategy

The commonly used action selection strategy in current reinforcement learning algorithms is the $\varepsilon - \text{greedy}$ strategy, which randomly selects and explores actions with a probability of ε when outputting actions through an action network, and selects the action with the highest Q value based on the action value function with a probability of $1-\varepsilon$. This random strategy can effectively learn the optimal strategy in simple environments, but in sparse reward environments such as formation control, it is difficult to obtain rewards through random strategies. At the same time, the algorithm converges slowly under random strategies, which does not meet the practical requirements of formation control. Therefore, a priority strategy is proposed, which is combined with the layered action space designed in the previous section, that is, wingmen prioritize certain actions based on observation states. The specific approach is to select the action library action_i based on the current observed state values, and the most important factor in this environment is the error between the y-direction and the expected distance. When the distance is far, choose the action library with a larger interval, and when the distance is close, choose the action library with a smaller interval. Then, prioritize the selection based on the probability of ε. When the position of the wingman is observed to be left, prioritize the right deviation action, i.e. $\psi > 0$, and vice versa. To accelerate algorithm convergence and learn the optimal strategy quickly.

4. Simulation verification based on PyTorch

4.1 Parameter Settings

The system employed Pytorch to construct a simulation environment for verification purposes. The DQN neural network utilized a three-layer convolutional neural network, where the activation function of each layer was the Relu function, and the optimization function was updated via the Adam algorithm. The remaining hyperparameters in the simulation program are presented in Table 2, encompassing parameters such as the iteration count and the learning rate.

Table. 2 Other parameter settings

parameter	value	parameter	value
epi_i	10000	$Maxstep$	200
$\psi_{\max}$	180	M	1e-6
$batchsize$	128	γ	0.95
ε	$1 \rightarrow 0.1$	τ	0.9
α	0.0001	$(\sigma_1, \sigma_2, \sigma_3)$	1
(μ_1, μ_2)	50	μ_3	20
$\tau_{\psi a}$	0.919	$\tau_{\psi b}$	0.919

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In Table 2, epi_i is the number of iterations; $Maxstep$ is the maximum number of steps, which is the maximum number of iterations in a single training/optimization process. Once this value is reached, the current task is stopped; $\psi_{\max}$ is the maximum threshold used to control the range of feature values or the upper limit of activation function output (such as the cutoff value of ReLU); $batchsize$ is the batch size; ε is the attenuation factor, representing the attenuation coefficient of the weight, gradually decreasing from 1 to 0.1 to achieve dynamic adjustment; α is penalty coefficient used as a penalty term in the loss function to control the impact of outliers or noise; (μ_1, μ_2) and μ_3 are the mean parameter, corresponding to the initialization of multimodal data or multi class features; γ is the learning rate; τ is the smoothing coefficient used for momentum optimization; $(\sigma_1, \sigma_2, \sigma_3)$ is the set of distributed parameters; $\tau_{\psi a}$ and $\tau_{\psi b}$ are weight parameters.

4.2 Simulation verification results

Figure 4 depicts the average loss curve acquired during the training procedure of heading angle control via a Deep Q-Network (DQN) algorithm. From a holistic viewpoint, the curve exhibits a distinct downward tendency, mirroring the model's learning advancement over time. In the initial stage of training, when the iteration number is relatively small, the average loss stays at a high level, nearly attaining 0.175. This implies that at the onset of the learning process, the model's predictions deviate significantly from the actual values, leading to a large error signal and sub - optimal policy performance.

As the training progresses and the iteration number rises, the average loss value decreases rapidly. By the time the iteration count nears 50, the average loss has already dropped below 0.025. This steep decline suggests that the model is effectively learning from interactions with the environment, gradually optimizing its decision - making strategy, and enhancing prediction precision. The substantial reduction in loss during this phase underscores the convergence ability of the DQN algorithm in this control task.

In the subsequent phase, from approximately 50 to 700 iterations, the average loss value stabilizes at a low level with only minor oscillations. Although slight variations exist, the overall trend remains stable, indicating that the model has reached a relatively stable performance state. During this period, the algorithm continues to conduct fine - tuning, yet no significant further improvement or deterioration is observed. This stabilization implies that the model has converged towards an effective policy, and additional training does not result in substantial alterations in loss, demonstrating the robustness and suitability of the DQN approach for this particular control problem.

This graph visually illustrates the fluctuation of the model's average loss value as the number of iterations increases during the training procedure of heading angle control via DQN. This phenomenon reflects the progression of the model from the initial learning phase to a state of gradual convergence and stable performance.

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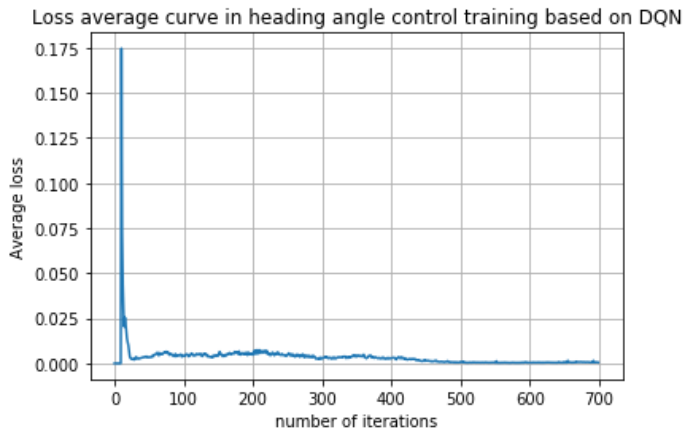

Figure. 4 Average loss curve in heading angle control training using the DQN

In the figure5, as the number of iterations approaches zero, the reward value rapidly declines to approximately -1750. Commencing from around 20 iterations, the reward value experiences a rapid increase and stabilizes at approximately 100 iterations, remaining at a level close to zero. At nearly 700 iterations, a slight decrease in the reward value is observed. Figure 5 illustrates the variation of the total reward value with the number

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of iterations during DQN-based control training, which reflects the learning and convergence processes during training.

The substantial upsurge commencing at approximately 20 iterations indicates that the DQN algorithm commences to acquire valuable patterns from the environment. It initiates the adjustment of control parameters in a manner that yields superior performance and consequently higher rewards. The stabilization of the reward value at around 100 iterations suggests that the algorithm has converged to a relatively stable control strategy. At this juncture, the system has learned a set of actions capable of sustaining a near - zero reward, which may signify an efficient and balanced operational state.

The marginal decline at nearly 700 iterations can be attributed to diverse factors. It is possible that the environment has undergone a minor alteration, or the algorithm has reached a local optimum and is beginning to explore other regions of the solution space. This slight decrease in the reward value might serve as an indication of the necessity to further optimize the training process, potentially through the adjustment of the learning rate or the exploration rate of the DQN algorithm.

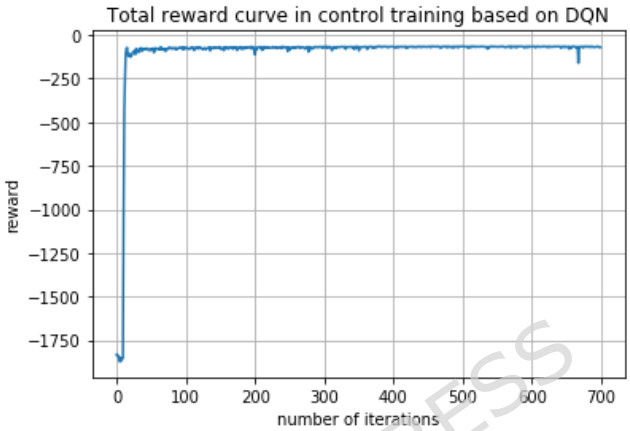

Figure. 5 Total reward curve within control training founded on DQN

As depicted in Figure 6, it is observable that under the control of the DQN, the yaw angle swiftly adjusts from the initial -80 degrees to 0 degrees and maintains stability subsequently. Figure 7 illustrates the rapid increase of the yaw angle from the initial -75 degrees to approximately 75 degrees under DQN control, followed by a process of stability maintenance. The DQN controllers in Figure 6 and Figure 7 can expeditiously and effectively adjust the yaw angle to the target value while ensuring the system's stability.

In addition to the adjustment of the yaw angle, the simulation also reveals the performance of other parameters under the control of the Deep Q-Network (DQN). For example, the rotational speed of the equipment exhibits a stable and predictable pattern. During the adjustment of the yaw angle, the rotational speed gradually rises from the initial value and subsequently stabilizes at an appropriate level. This suggests that the DQN controller is not only capable of handling the yaw angle adjustment but also coordinating with other parameters to guarantee the overall stability and efficiency of the system.

Overall, the simulation verification results illustrate the outstanding performance of the DQN controller in adjusting the yaw angle and coordinating other system parameters. It can ensure the stability and efficiency of the system while adapting effectively to different initial conditions. These results offer robust support for the practical application of the DQN controller in relevant fields.

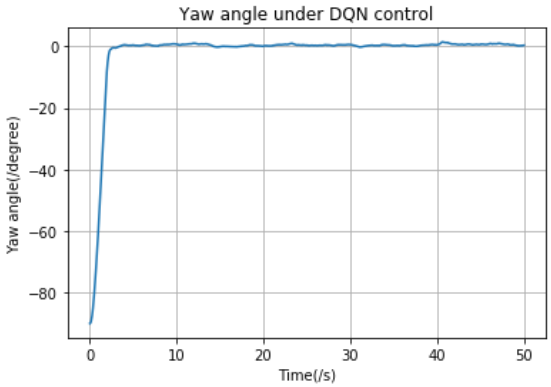

Figure. 6 Yaw angle under the control of DQN: ranging from -80 degrees to 0 degrees

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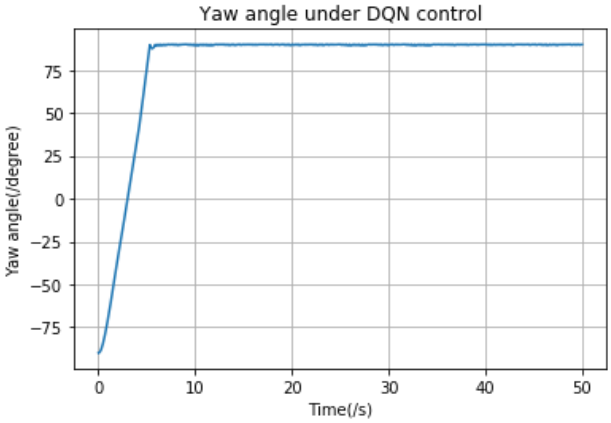

Figure. 7 Yaw angle under the control of DQN: ranges from -75 degrees to nearly 75 degrees.

Figure 8 depicts the alterations in the generated control variables at diverse time points. Noticeably, significant fluctuations, including both sharp increases and decreases, are evident at multiple time points, which illustrates the temporal dynamic characteristics of the control variables. These variations might be associated with the experimental procedure.

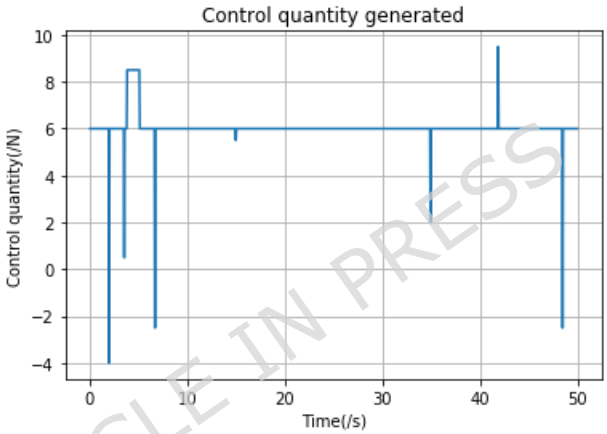

Figure. 8 Regulate the quantity produced

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As depicted in Figure 9, it is evident that as time approaches 0 seconds, the adjustment speed of the yaw angle gradually decelerates and converges towards a value of 0°/s. Precisely commencing from the 0 - second point, the speed experiences a rapid and significant upsurge, reaching its peak magnitude within an approximate time span of 2 to 3 seconds. During this initial acceleration stage, the speed exceeds 40°/s, signifying a highly dynamic reaction. After attaining this maximum value, the speed initiates a rapid decline and subsequently enters a phase of oscillations, with the speed fluctuating between 30°/s and 40°/s. As time progresses, these oscillations gradually attenuate, and the speed trends towards a more stable state.

From 20 seconds to 40 seconds, the adjustment speed remains relatively constant, maintaining an average value of approximately 35°/s, with only minor and sporadic fluctuations discernible during this period. As the time nears the 50 - second mark, there are again slight variations in speed; however, the amplitude of these disturbances is relatively small, and the overall speed continues to remain stable at around 35°/s.

Overall, Figure 9 visually portrays the dynamic temporal evolution of the yaw angle adjustment speed under DQN control, highlighting an initial phase characterized by intense and rapid changes that ultimately transitions into a more stable and consistent behavioral pattern over the observed time frame.

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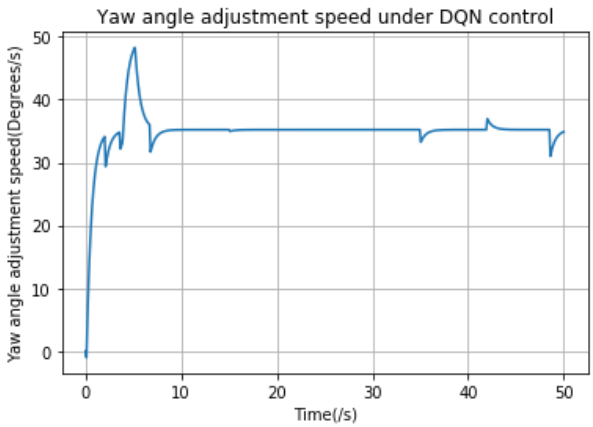

Figure. 9 Yaw angle adjustment velocity under the control of DQN

In the initial stage (approximately 0 - 5 seconds), the reward value in Figure 10 demonstrated notable fluctuations. The value initially ascended rapidly to a relatively high level near - 75, and subsequently declined swiftly to a lower level close to - 200. This phenomenon might imply that during the early operation of the system, the agent was in a state of exploration and environmental adaptation, and its decision - making behavior had not yet established stable patterns, leading to substantial fluctuations in the reward value.

In the intermediate stage (approximately 5 - 10 seconds), the reward value underwent a short - term rebound subsequent to its initial decline, accompanied by a certain degree of fluctuation. These fluctuations could reflect the system's attempts to explore diverse strategies or actions in pursuit of a more optimal decision - making path. As a result, despite the overall adjustment period, the reward value still manifested a certain level of instability at specific moments.

In the stabilization stage (approximately 10 - 50 seconds), the reward value gradually stabilized and remained at a relatively low level, fluctuating around - 225. This indicates that after the initial exploration and learning phase, the DQN control strategy has attained a relatively stable state. The system's decision - making behavior becomes consistent, and the reward value no longer exhibits significant fluctuations, suggesting that the agent has converged to a relatively optimal strategy to a certain extent.

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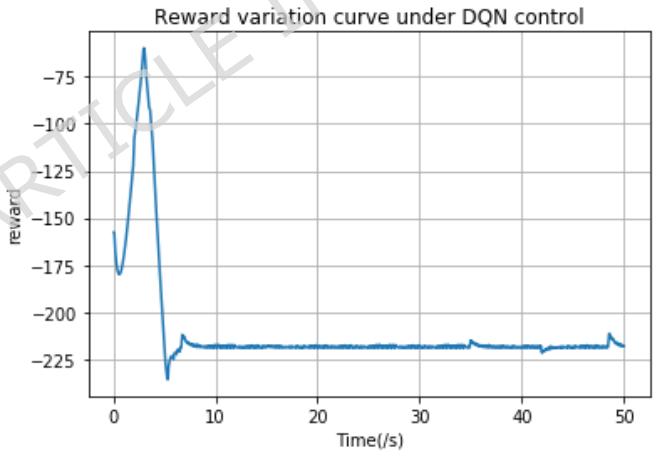

Figure. 10 Reward Variation Curve under DQN Control

Overall, under the control of the DQN, the reward value undergoes significant fluctuations initially and then gradually stabilizes over time. This phenomenon reflects the transition of the DQN algorithm within the system from the exploration phase to the relatively stable strategy execution phase.

The Figure 11 shows the trajectory comparison of the Leader Follower Formation. The horizontal axis of the chart represents control time, and the vertical axis represents distance error. In Figure 11, a comparison was made between the PID control method, DDPG control method, and DQN control method in terms of the performance of followers during the leader following process. These three control methods enable followers to roughly track the leader's path. However, at a specific stage, followers using DQN control method exhibit tracking performance closer to that of the leader, indicating its advantage in path following.

In the comparison of PID control approaches, the DQN control approach exhibits greater adaptability and precision when addressing complex paths and dynamic variations. Although the PID control approach is straightforward and easy to implement, its performance is frequently restricted when dealing with nonlinear and

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time-varying systems. Conversely, the DQN control approach can automatically learn and optimize control strategies via deep learning and reinforcement learning techniques, thereby attaining more accurate path tracking in complex environments. In the comparison with DDPG, the DQN effectively mitigates training instability and overestimation issues through techniques such as Experience Replay and Target Network, demonstrating favorable convergence and stability. It can handle high-dimensional state spaces, and the training process is relatively controllable. In contrast, although DDPG performs remarkably well in continuous action spaces, the simplicity and maturity of DQN render it a more efficient option in discrete action scenarios. It should be noted that for DQN, although discrete actions may lead to control jitter, a hierarchical action space can effectively mitigate this issue. The advantages of choosing DQN are training stability and algorithm simplicity. The DQN architecture is more lightweight than RL algorithms such as DDPG, with lower memory consumption and latency in embedded unmanned aerial vehicle systems. For the formation flight control of unmanned aerial vehicles, the DQN algorithm is sufficient to meet the formation control requirements of unmanned aerial vehicles.

In conclusion, the DQN control method has exhibited substantial advantages in the path tracking task of the Leader - Follower Formation. Its robust learning capability and adaptability empower it to attain more precise and stable path tracking in intricate environments, offering robust support for practical applications.

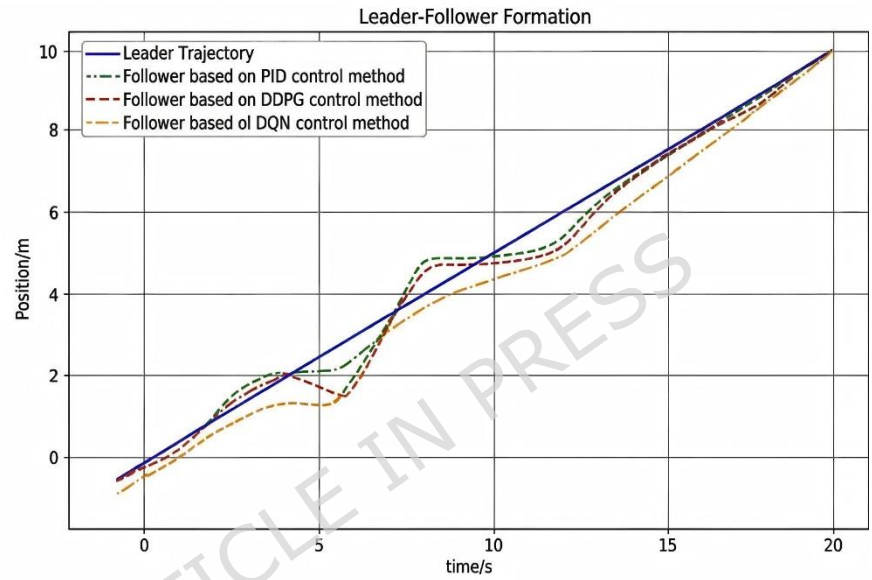

Figure. 11 Leader-Follower Formation

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6. Conclusion

In light of the limited intelligence level in drone formation flight and the high demand for model information in controller design, a formation controller based on DQN is devised. Via simulation experiments, it is ultimately demonstrated that precise model information is not requisite in the design of the DQN controller. Through training, the wingman can autonomously acquire the optimal strategy for following the leader, as well as tracking and maintaining the desired distance. By conducting a comparison with the PID control method, it is concluded that the DQN control method exhibits a swifter convergence speed and superior control efficacy.

Moreover, the DQN based formation controller shows strong adaptability to different scenarios. It can effectively handle various complex situations during the drone formation flight, such as sudden changes in the leader's trajectory or environmental disturbances. This adaptability is mainly due to the self - learning ability of the DQN algorithm, which allows the controller to continuously adjust and optimize its strategy according to the actual situation.

In addition, the research results also provide valuable insights for the further development of drone formation flight technology. The successful application of the DQN algorithm in this field opens up a new direction for the design of more intelligent and efficient formation controllers. Future research can focus on improving the performance of the DQN based controller in more extreme and dynamic environments, as well as exploring the combination of the DQN algorithm with other advanced technologies to further enhance the overall capabilities of drone formation flight systems.

Data sharing agreement

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request. All relevant data are within the manuscript. All figures and tables are made by the author.

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Declaration of Conflicting Interests

The author declared no financial and non-financial conflicting interests. And the author declared no potential conflicts of interest with respect to the research, author-ship, and publication of this article.

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Ethical approval

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Consent to participate

Not applicable

Consent to publish

Not applicable

Clinical trial number

Not applicable.

Author Contribution

Zhenqi He contributed to the conception of the study, wrote the first draft of the manuscript and worked on the coding of tables and figures and approved the final manuscript.

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